Hamdy Khaled

Software Developer | Full-Stack .NET Developer | C# | ASP.NET Core | React | TypeScript
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Professional Summary

Software Developer (Full-Stack .NET) with hands-on experience in backend development, frontend development, web development, and RESTful API design using C#, .NET, ASP.NET Core Web API, Entity Framework Core, SQL Server, React, and TypeScript. Built two full-stack applications: a real-time SaaS platform with 80+ API endpoints and a multi-vendor e-commerce platform with 170+ endpoints, applying Clean Architecture, CQRS, three-tier architecture, SOLID principles, and design patterns. Strong in authentication and authorization (JWT, OAuth, role-based access control, resource-based access control), Redis caching, SignalR, Payment integration, database design, and performance optimization. Experienced with Agile/Scrum, Git, and GitHub.

Projects

Task and Workspace Management Platform | Full-Stack SaaS (Clean Architecture + CQRS)

GitHub: Backend | Frontend

Tech Stack: ASP.NET Core 10, C#, MediatR, EF Core, SQL Server, Redis, SignalR, JWT, Google OAuth, QuestPDF, React 19, TypeScript, Vite, Redux Toolkit, TanStack Query, Tailwind CSS

Designed and developed a full-stack team collaboration SaaS web application with an ASP.NET Core 10 REST API (13 controllers, 80+ RESTful endpoints, 12 entities) and a React 19 + TypeScript single-page application.

- Architected a 4-layer Clean Architecture + CQRS backend with MediatR, a FluentValidation pipeline behavior, ErrorOr results, and RFC 7807 problem details across 80+ API endpoints.
- Implemented secure authentication and authorization with ASP.NET Identity and JWT in HttpOnly cookies (20-minute access tokens, 30-day rotating refresh tokens with revocation), Google OAuth, OTP password reset, and email-change confirmation.
- Implemented two-level role-based access control (RBAC): global Admin/User and workspace Owner/Project Manager/Member, with 3 custom authorization policies and workspace invitations that expire after 60 days.
- Delivered real-time notifications with SignalR across 7 event types, persisted read/unread state, and automatic reconnect synced to the UI through TanStack Query cache invalidation.
- Developed admin and user dashboards plus 9 report types (including 2 QuestPDF PDF exports), with Redis caching of 5 minutes for dashboards and 10 minutes for reports.
- Optimized SQL Server performance with EF Core and LINQ: 8 composite, filtered, and unique indexes, no-tracking reads, projection queries, soft-delete query filters, and server-side pagination on list endpoints.
- Implemented URL-based API versioning with Asp.Versioning.Mvc, so endpoints can evolve without breaking existing clients.
- Hardened the API with 8 fixed-window rate-limiting policies (HTTP 429), 5 background MailKit email queues, a 30-minute unconfirmed-account cleanup background job, global exception handling, and Serilog + Seq logging.
- Mapped unique and foreign-key database violations to clear HTTP 409 responses through Unit of Work transactions, keeping API error handling consistent and predictable.
- Engineered frontend state management with TanStack Query (70+ custom hooks, infinite scroll) and Redux Toolkit; built an Axios interceptor that refreshes the token once and replays queued requests after a 401.
- Built a responsive React 19 UI with 36 pages, 3 route guards, Chart.js analytics, Formik + Yup form validation, Framer Motion animations, dark mode, and English/Arabic RTL support via i18next.

Amazon E-Commerce Platform | Full-Stack Multi-Vendor Marketplace

GitHub: Backend | Frontend

Tech Stack: ASP.NET Core 10, C#, React, TypeScript, SQL Server, Entity Framework Core, Redis, Stripe

Developed a multi-vendor e-commerce marketplace backend (25 controllers, 170+ RESTful endpoints, 27 entities, 61 EF Core migrations) on a three-layer architecture with Generic Repository and Unit of Work.

- Structured the backend into 3 layers with 35 services, 28 repositories behind a Unit of Work, and 59 DTOs mapped through 30 AutoMapper profiles, keeping controllers thin and business logic in one layer.
- Implemented authentication and authorization with signed and encrypted JWT, database-backed refresh tokens in HttpOnly cookies, 10-minute OTP verification, Google/GitHub OAuth, and 4-role RBAC (Admin, Customer, Seller, Delivery Agent).

- Integrated Stripe Checkout payment processing with 5 signature-verified webhook events, cash on delivery, and automatic refunds; made checkout retry-safe with Redis idempotency keys and guarded SQL stock updates.
- Modeled a 4-stage order lifecycle (processing, shipped, delivered, canceled) plus returns as append-only status history, with delivery-agent assignment, stock restoration, and queued status emails.
- Used Redis caching for city data, recent searches (last 10 per user), and checkout idempotency keys; added a 12-hour background cache refresh and per-IP rate limiting (20 requests per 10 seconds).
- Developed bilingual Arabic/English search with automatic language detection, Cloudinary multi-image uploads, and verified-purchase reviews enforced by a unique database index.
- Built a responsive React + TypeScript storefront with 24 pages (Redux Toolkit, TanStack Query, Tailwind CSS, Axios) covering search, cart, checkout, order tracking, RTL support, and dark mode.

Technical Skills

Languages: C#, TypeScript, JavaScript (ES6+), SQL, T-SQL, HTML5, CSS3

Backend: .NET, ASP.NET Core Web API, ASP.NET Identity, Entity Framework Core (ORM), LINQ, RESTful APIs, REST API Design, JWT Authentication, OAuth 2.0 (Google, GitHub), Authentication and Authorization, Role-Based Access Control (RBAC), SignalR, MediatR, AutoMapper, Mapster, FluentValidation, ErrorOr, Serilog, MailKit, QuestPDF, Swagger (OpenAPI)

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, Bootstrap, Material UI (MUI), Redux Toolkit, TanStack Query (React Query), React Router, Axios, Formik, Yup, Chart.js, i18next, Framer Motion, Vite, Responsive Design

Databases and Caching: SQL Server, T-SQL, SQL Server Management Studio (SSMS), Redis, Database Design, Query Optimization, Indexing, Entity Framework Migrations

Architecture and Concepts: Clean Architecture, CQRS, Layered Architecture, Design Patterns, SOLID Principles, Object-Oriented Programming (OOP), Data Structures and Algorithms, Repository Pattern, Unit of Work, Mediator Pattern, Dependency Injection, Background Services, Caching, Rate Limiting, Real-Time Communication, Idempotency, Performance Optimization

Integrations: Payment Gateways (e.g., Stripe), OAuth 2.0 Providers (e.g., Google, GitHub), Cloudinary, SMTP Email

Methodologies and Tools: Agile/Scrum, Software Development Life Cycle (SDLC), CI/CD Concepts (GitHub, Automated Deployments), Git, GitHub, Postman, Swagger, Visual Studio, VS Code, Jira, Slack

Certifications

- **Programming Advices (Online Courses): Programming Fundamentals Roadmap.** 25 certificates covering programming fundamentals, algorithms and data structures, object-oriented programming (OOP), C#, SQL and database engineering, and RESTful APIs.
- **Udemy:** The Complete JavaScript Course (Jonas Schmedtmann).

Education

Bachelor's Degree in Social Work | Helwan University | Cairo, Egypt | 2022 - 2026